WHERE CLAUSE

----> WHERE CLAUSE KYA HAI?? :- WHERE CLAUSE QUERY ME APNE SATH CONDITION LETA HAI
----> WHERE CLAUSE KISKE SATH LIKHTE HAI?:- WHERE CLAUSE LIKHNE KE BOHOT SARE WAYS HAI

1. COMPARISON OPERATORS :- =, !=, >, <, <=, >=

--> INT PE HI KAM KREGA VARCHAR ME NAHI KRTA HAI BUT WO DIFFERNT CASE HO JATA HAI

```
SELECT * FROM users
WHERE AGE > 18 ;
```

```
SELECT * FROM users
WHERE NAME> 'G' ;
```

--> YAHA PEHLA TO THIK SE HI CHALEGA BUT DUSRI QUERY ME OUTPUT ME JO NAME G KE BAAD KE LETTERS HOGE WO AAYENGE

2. LOGICAL OPERATORS :- AND, OR, NOT

--> MULTIPLE FILTER EK SATH LAGANE DETE HAI

```
SELECT * FROM users
WHERE age > 18 AND city = 'Delhi' ;
```

3. BETWEEN (RANGE FILTER)

--> YE KOI BHI CHIZ KI RANGE FIND KRTA HAI JESE SALARY FIND KRNI HO 100000 TO 200000 KE BICH ME TO ISKA USE HOTA HAI

```
SELECT * FROM users
WHERE salary BETWEEN 100000 AND 200000;
```

4. IN (MULTIPLE VALUES) & NOT IN

--> YE FILTER CHOOSE KRNE KE LIYE MULTIPLE OPTIONS PROVIDE KRTA HAI

```
SELECT * FROM users
WHERE city IN ('Delhi', 'Mumbai', 'Pune');
```

5. LIKE (PATTERN MATCHING)

```
SELECT * FROM users
WHERE name LIKE 'P%';    -- starts with P

SELECT * FROM users
WHERE name LIKE '%t';    -- ends with t

SELECT * FROM users
WHERE name LIKE '_r%';   -- second letter r
```

6. IS NULL / IS NOT NULL

--> YE NULL OR NOT NULL VALUE KO FIND KRNE KA FILTER HAI

```
SELECT * FROM users
WHERE phone_number IS NULL;
```

7. EXISTS / NOT EXIST

--> YE SUBQUERY KE SATH WORK KRTA HAI KI WO DIKHAO JO DATA EXIST HAI YA NAHI

```
SELECT name FROM users u
WHERE EXISTS(
        SELECT 1 FROM orders o
        WHERE o.user_id = u.id
);
```

--> YE EXAMPLE ME ESA HAI KI KONSA ORDER AYA HAI WO userDIKHAO

DIFFERENCE

Feature	IN	EXISTS
Working	full list compare	row by row check
Speed	slow (large data)	faster (stop early)
Use case	small data	large data

DISTINCT

-->YE DUPLICATE VALUES HATA TA HAI OR FIR OUT PUT DEEGA MTLB UNIQUE VALUE HI DEGA
-->IT MEANS AGAR CITY KI COLUMN AAP SELET KR RHE HO OR DISTINCT CITY LIKHA TO YE SIRF DIFF CITY NAMES HI DEGA

name	city
Prit	Rajkot
Rahul	Ahmedabad
Amit	Rajkot

<-- YE TABLE HAI JISME CITY RAJKOT 2 BAAR HAI OR AHMD HAI TO DISTICNT CITY LIKHNE SE SIRF DIFF CITY HI AYEGI

```
SELECT  DISTINCT CITY FROM USERS;
```

OUTPUT:--->

city
Rajkot
Ahmedabad

NOTE :- DISTICNT EXPENSIVE HOTA HAI SO LARGE DATA ME SLOW HOGA

ORDER BY

--> ORDER BY DATA KO ASCENDING OR DESECING ORDER ME SORT KARTA HAI.
--> JESE KI AGAR HAME AGE YA KOI NAME HO UNHE LINES ME CHHIYE LIKE PROPER ORDER ME TAB USE HOTA HAI

```
SELECT * FROM users
ORDER BY age DESC;
```

OUTPUT-->

Output

name	age
Rahul	25
Prit	22
Amit	19

LIMIT

--> YE OUTPUT ME KITNI ROWS HME CHHAIYE WO LIMIT SE DEFINE KR SKTE HAI HAM
--> LARGE DATA ME HAME SAB DETA TO CHHIYE NAHI HOTE TO SIRF UPER KE DATA CHHIAYE TO LIMIT LAGA KR LE SKTE HAI

```
SELECT name FROM users
LIMIT 2 ;
```

Output

name
Prit
Rahul

GROUP BY

-->YE KISI BHI CHIZ KA GROUP BANATA HAI ORR VALUES KO GROUP ME DIKHAYEGA
-->JEEE DIFF TYPE OF CITY HAI TO CITY KA GROUP BANAYEGA JO SAME HAI USKA

```
SELECT city, COUNT(*)
FROM users
GROUP BY city;
```

Output

city	count
Rajkot	2
Ahmedabad	1

HAVING

-->HAVING CONDITION WHERE KE JESI HI HAI DONO ME FILTER HI LAGTA HAI BUT HAVING SIRF GROUP BY PE FILTER LAGATA HAI
-->WHERE SAB ME FILTER LAGAYEGA HAVING GROUP BY KO FILTER KREGA FIR VAKUE DEGA

```
SELECT city, COUNT(*)
FROM users
GROUP BY city
HAVING COUNT(*) > 1;
```

Output

city
Rajkot

--> KYUKI YAHA PE RAJKOT 2 BAAR AAYA HAI OR HMNE HAVING COUNT(8)>1 RAKHA HAI SO AHMD NAHI AAYEGA ISME

AGGREGATE FUNCTION

COUNT ----------> SELECT COUNT(*) FROM users ;

SUM ----------> SELECT SUM(salary) FROM users ;

AVG ----------> SELECT AVG(age) FROM users ;

MIN ----------> SELECT MIN(age) FROM users ;

MAX ----------> SELECT MAX(age) FROM users ;

SUBQUERY

SINGLE ROW

--SYNTEX:-

```
SELECT *
FROM Employees
WHERE dept_id = (
   SELECT dept_id FROM Departments WHERE dept_name = 'HR'
);
```

MULTIPLE ROWS

--SYNTEX:-

```
SELECT *
FROM Employees
WHERE dept_id IN (
   SELECT dept_id FROM Departments
);
```

CORRELATED

--SYNTEX:-

```
SELECT e.*
FROM Employees e
WHERE EXISTS (
   SELECT 1
   FROM Departments d
   WHERE e.dept_id = d.dept_id
);
```

AS(ALIAS)

TABLE NAME

--SYNTEX:-

```
SELECT e.name, d.dept_name
FROM Employees e
JOIN Departments d
ON e.dept_id = d.dept_id;
```

COLUMN NAME

--SYNTEX:-

```
SELECT name AS employee_name
FROM Employees;
```

CASE STATEMENT

----> YE SQL KA IF ELSE HAIN

-->SYNTEX:-

```
SELECT name,
CASE
   WHEN dept_id = 101 THEN 'HR'
   WHEN dept_id = 102 THEN 'IT'
   ELSE 'Unknown'
END AS dept_category
FROM Employees;
```

SET OPERATIONS

1. UNION :-

-- UNION DO QUERY KE RESULT KO COMBINED KRTA HAI OR DUPLICATE REMOVE KRTA HAI

SYNTEX:-

```
SELECT name FROM Employees
UNION
SELECT name FROM Managers;
```

2. UNION ALL :-

-- UNION ALL DO QUERY KE RESULT KO COMBINED KRTA HAI OR DUPLICATE BHI LLOW KRTA HAI

SYNTEX:-

```
SELECT name FROM Employees
UNION ALL
SELECT name FROM Managers;
```

3. INTERSECT :-

-- DONO QUERY MESE SIRF COMMON VALUE HI DETA HAI

SYNTEX:-

```
SELECT name FROM Employees
INTERSECT
SELECT name FROM Managers;
```

3. EXCEPT :-

-- WO VALUE JO PEHLI QUERY ME HAI BUT DUSRI ME NAHI WO DIKHAYEGA

SYNTEX:-

```
SELECT name FROM Employees
EXCEPT
SELECT name FROM Managers;
```